FBS Cell Biology — NBME Study Guide

Organelles · Plasma membrane · Cytoskeleton · Cell cycle · Apoptosis (includes Anki-deck cell biology)

Red = extra high yield (classic NBME) Black = core content *Italic blue = mnemonic*

1. Organelles

- **Cytosol** = fluid portion only; **cytoplasm** = cytosol + organelles + inclusions (everything inside the plasma membrane except the nucleus).

Nucleus

- Double membrane (outer continuous with RER), **nuclear pores** (bidirectional transport; mRNA export, protein import via NLS). Blocking pores → mRNA can't reach ribosomes → no translation.
- **Nuclear lamina** (lamins A, B, C — type V intermediate filaments) maintains shape; LINC complexes connect it to the cytoskeleton. **Lamin A mutation → progeria (Hutchinson-Gilford).**
- **Nucleolus** = site of **rRNA synthesis (RNA polymerase I)** and ribosomal subunit assembly. RNA Pol II = mRNA; Pol III = tRNA, 5S rRNA.
- mRNA path: transcription (nucleus) → capping, splicing, poly-A tail → export via pores → translation on ribosomes.
- Heterochromatin = condensed, inactive (Barr body); euchromatin = open, active.

Ribosomes, RER, Golgi

- Eukaryotic 80S = 60S + 40S (prokaryotic 70S = 50S + 30S → antibiotic targets).
- **Signal sequence + SRP decide fate**: SRP binds the N-terminal signal peptide, pauses translation, docks the ribosome on the RER (SRP receptor) → protein enters ER lumen via translocon (Sec61).
- **RER makes secretory, membrane and lysosomal proteins** (and N-linked glycosylation begins). **Free ribosomes** make cytosolic, nuclear, mitochondrial and **peroxisomal** proteins.
- Cells rich in RER: plasma cells, pancreatic acinar cells, goblet cells; Nissl bodies in neurons.
- **SER: steroid hormone synthesis (adrenal cortex, Leydig cells, gonads), lipid synthesis, detoxification (hepatocyte cytochrome P450), glycogen breakdown (glucose-6-phosphatase), Ca^{2+} storage (sarcoplasmic reticulum in muscle).**
- **Golgi**: cis (receives from ER) → trans (ships out). Modifies N-glycans, adds **O-linked sugars** (Ser/Thr), sulfation of GAGs, proteolytic processing, **adds mannose-6-phosphate (M6P) to lysosomal enzymes**.

Coat	Direction
COPII	**Anterograde: ER → cis-Golgi** ***("2 = forward to Golgi")***
COPI	**Retrograde: Golgi → ER (and intra-Golgi), KDEL retrieval**
Clathrin	**trans-Golgi → lysosomes; plasma membrane → endosomes (receptor-mediated endocytosis, e.g., LDL)**

Insulin processing (Anki): preproinsulin (RER, signal peptide cleaved) → proinsulin (folds, 3 disulfides) → Golgi/secretory granules: prohormone convertases cut out **C-peptide** → insulin + C-peptide secreted 1:1. **C-peptide** distinguishes endogenous insulin (insulinoma, sulfonylureas: ↑ C-peptide) from exogenous insulin (factitious: ↓ C-peptide). ↑ proinsulin → defective convertase processing.

Lysosomes

- Acid hydrolases (pH ≈ 4.5–5, H^+-ATPase); digest by endocytosis, phagocytosis, autophagy.
- **Trafficking: RER synthesis → cis-Golgi GlcNAc-phosphotransferase adds M6P → M6P receptor in trans-Golgi → clathrin vesicle → late endosome → lysosome.**
- **I-cell disease (mucolipidosis II)**: defective GlcNAc-phosphotransferase → no M6P tag → enzymes secreted → **↑ plasma lysosomal enzymes**, inclusion bodies, coarse facies, gingival hyperplasia, restricted joints, clouded corneas; AR, fatal in childhood.
- Classic storage diseases (Tay–Sachs, Gaucher, Niemann–Pick, Fabry, Hurler, Hunter, metachromatic leukodystrophy) = a **single enzyme** deficient; I-cell = **all** enzymes mis-sorted.

Disease	Enzyme	Key feature

Tay–Sachs	**Hexosaminidase A (GM2)**	**Cherry-red macula, no hepatosplenomegaly, onion-skin lysosomes**
Gaucher	**Glucocerebrosidase**	**Most common; crumpled tissue-paper macrophages, bone crises, HSM**
Niemann–Pick	Sphingomyelinase	Cherry-red macula + HSM, foamy cells
Fabry (XR)	α-galactosidase A	Neuropathic pain, angiokeratomas, renal/cardiac disease
Hurler / Hunter (XR)	α-L-iduronidase / iduronate sulfatase	GAG accumulation; corneal clouding in Hurler only *(Hunters aim for the X)*
Metachromatic leukodystrophy	Arylsulfatase A	Demyelination

Mitochondria

- Double membrane; inner membrane folds = **cristae** (↑ surface for ETC/ATP synthase). Matrix: TCA cycle, β-oxidation, own circular DNA.
- **mtDNA: ~37 genes, maternal inheritance, heteroplasmy** (mixed normal/mutant copies) → variable severity between siblings and tissues; threshold effect.
- Endosymbiotic evidence: own circular DNA, 70S-like ribosomes, binary fission, double membrane, cardiolipin.
- **Mitochondrial myopathy: ragged-red fibers** (Gomori trichrome); lactic acidosis; tissues with high ATP demand (brain, muscle, eye) affected.
- MELAS (encephalomyopathy, lactic acidosis, stroke-like episodes), MERRF (myoclonic epilepsy, ragged-red fibers), **LHON (Leber hereditary optic neuropathy — sudden bilateral vision loss in young men; affected males cannot pass it on)**.
- Apoptosis: **cytochrome c release** (outer membrane permeabilized by Bax/Bak) → apoptosome.

Peroxisomes

- Single membrane; proteins made on **free ribosomes**, imported via PTS1 (C-terminal SKL) and **peroxins (PEX)**; replicate by fission; no DNA.
- **Functions: β-oxidation of very-long-chain fatty acids (VLCFA) and branched-chain (α-oxidation of phytanic acid), plasmalogen synthesis (myelin), bile acid synthesis, H_2O_2 production + catalase breakdown.**
- Peroxisomal β-oxidation makes heat + H_2O_2 (no ATP); shortened chains go to mitochondria to finish.
- **Zellweger syndrome**: PEX gene defect → no functional peroxisomes; hypotonia, seizures, hepatomegaly, craniofacial anomalies, early death.
- **X-linked adrenoleukodystrophy**: ABCD1 transporter defect → VLCFA buildup → adrenal insufficiency + progressive demyelination in boys.
- **Refsum disease**: phytanic acid accumulation (α-oxidation defect) → ataxia, retinitis pigmentosa, ichthyosis, neuropathy; restrict dietary phytanic acid (dairy, ruminant fat).

2. Plasma membrane

- Functions: barrier, transport, signaling, cell recognition/adhesion. Basic structure = **phospholipid bilayer** (amphipathic: polar heads, nonpolar tails; held by hydrophobic interactions).
- **Outer leaflet: phosphatidylcholine, sphingomyelin, glycolipids. Inner leaflet: phosphatidylserine (PS), phosphatidylethanolamine, phosphatidylinositol.** PS flipping to the outside = "eat me" signal in apoptosis (annexin V detects it).
- **Cholesterol = fluidity buffer**: at high temperature ↓ fluidity (restrains tails); at low temperature ↑ fluidity (prevents packing).
- Unsaturated (kinked) fatty acids & short chains ↑ fluidity; saturated & long chains and cold ↓ fluidity (hypothermia → stiff membranes).
- **Glycolipids**: sugars face outside; recognition. **Cholera toxin binds ganglioside GM1** → ↑ cAMP → secretory diarrhea.
- **Proteins**: integral (transmembrane α-helices or β-barrels; channels, carriers, receptors), peripheral (non-covalent, e.g., spectrin, cytochrome c), lipid-anchored (Ras via prenylation; GPI-anchored proteins).
- **ABO blood type**: glycosyltransferases add sugars to H antigen — A adds GalNAc, B adds galactose, **O has no active transferase** (bare H antigen). Bombay phenotype lacks H.

- **ABC transporters (P-glycoprotein/MDR1)** pump drugs out using ATP → chemotherapy resistance; at the blood–brain barrier they keep many drugs out of the CNS.
- **Glycocalyx**: carbohydrate coat — protection, recognition, endothelial anti-adhesion, enterocyte brush-border enzymes.
- **Lipid rafts**: cholesterol + sphingolipid-rich, less fluid microdomains for signaling. **Caveolae** (caveolin) = flask-shaped rafts; endocytosis/transcytosis, abundant in endothelium, adipocytes, smooth muscle; absent in neurons/RBCs.
- **Polarity**: apical vs basolateral domains separated by **tight junctions (claudins, occludin)**; classic example = intestinal/renal epithelium.

3. Cytoskeleton

	Microfilaments (actin)	Intermediate filaments	Microtubules
Diameter	~7 nm (smallest)	~10 nm	~25 nm (largest, hollow)
Subunit	G-actin → F-actin	Tissue-specific proteins	α/β-tubulin dimers, 13 protofilaments
Energy	**ATP; treadmilling**	None; very stable	**GTP; dynamic instability**
Polarity	Yes (+ barbed)	No	Yes (− at centrosome, + at periphery)
Functions	Microvilli, contractile ring, cell motility, adherens junctions	**Tensile strength; tumor markers**	Mitotic spindle, cilia/flagella, axonal transport
Motors	Myosin	—	**Kinesin (+ end, anterograde), dynein (− end, retrograde)**

- **Actin drugs**: cytochalasin (blocks polymerization), phalloidin (stabilizes).
- **Spectrin–ankyrin–band 3–protein 4.1** = RBC membrane skeleton. **Hereditary spherocytosis** (usually AD): spherocytes without central pallor, ↑ MCHC, splenomegaly, hemolytic anemia, gallstones; **osmotic fragility / EMA binding test**; splenectomy.
- **Dystrophin** links actin to the dystroglycan complex/ECM in muscle. **Duchenne** (X-linked, frameshift deletion → no dystrophin): proximal weakness by age ~5, **Gowers sign**, calf pseudohypertrophy, ↑ CK, dilated cardiomyopathy. **Becker**: in-frame deletion → truncated dystrophin, milder, later onset.

Intermediate filament	Tissue / tumor marker
Cytokeratin	**Epithelium → carcinomas**
Vimentin	**Mesenchyme (fibroblasts, endothelium) → sarcomas**
Desmin	**Muscle → rhabdo/leiomyosarcoma**
GFAP	**Astrocytes/glia → gliomas**
Neurofilaments	Neurons → neuronal tumors
Lamins	Nuclear envelope (all cells)

- **Microtubule drugs**: **vincristine/vinblastine** block polymerization; **paclitaxel (taxanes)** stabilize (block depolymerization) *("Taxes stabilize society")*; **colchicine** blocks polymerization (gout, ↓ neutrophil migration); griseofulvin; mebendazole. All arrest mitosis (M phase). **Vincristine → peripheral neuropathy** (axonal transport depends on microtubules); vinblastine → bone-marrow suppression.
- Kinesin/dynein: rabies, HSV, polio and tetanus toxin travel **retrograde via dynein** to the cell body.
- **Cilia: 9+2 axoneme** (motile: airway, fallopian tube, ependyma) with **dynein arms**; basal body = 9 triplets; primary (nonmotile, 9+0) cilia sense (kidney tubule flow, photoreceptors, embryonic node). Ciliopathies: polycystic kidney disease, Bardet–Biedl.
- **Kartagener syndrome (primary ciliary dyskinesia)**: AR dynein arm defect → infertility (immotile sperm; ↑ ectopic pregnancy), recurrent sinusitis, bronchiectasis, **situs inversus** (~50%); ↓ nasal NO.
- **Centrosome** (pair of centrioles, γ-tubulin) = MTOC; organizes spindle. Kinetochore microtubules attach to chromosomes; polar/interpolar push poles apart; astral anchor to cortex.

4. Cell cycle

- Phases: **G1** (growth; variable length) → **S** (DNA replication: DNA content doubles 2c → 4c, still 2n) → **G2** (prep, check DNA) → **M** (prophase, prometaphase, metaphase, anaphase, telophase) → cytokinesis.

- **G0** = quiescent, out of cycle. Cell types: **labile** (never G0: skin, gut epithelium, bone marrow, hair follicles), **stable/quiescent** (G0 but can re-enter: hepatocytes, kidney tubule cells, fibroblasts), **permanent** (cannot divide: neurons, skeletal & cardiac muscle, RBCs).
- **Regulation**: **cyclin–CDK complexes** drive transitions (cyclin D–CDK4/6 in G1, cyclin E–CDK2 G1/S, cyclin A S, cyclin B–CDK1 = MPF for G2/M). CDKs are constitutive; cyclins rise and fall (ubiquitin–proteasome degradation; APC/C at metaphase→anaphase).
- **Rb** (tumor suppressor) binds and inhibits **E2F**; cyclin D–CDK4 **phosphorylates Rb** → releases E2F → G1→S. **HPV E7 binds Rb; E6 degrades p53.**
- **p53 ("guardian of the genome")**: DNA damage → p53 → **p21** (CDK inhibitor) → G1 arrest for repair; if irreparable → apoptosis (↑ Bax/PUMA/NOXA). Mutated in >50% of cancers; Li–Fraumeni syndrome (germline).
- Checkpoints: G1/S (DNA damage), G2/M (replication complete, DNA intact), **spindle (M) checkpoint** (all kinetochores attached) — failure → aneuploidy.
- **Mitosis**: 1 diploid → 2 identical diploid cells. **Meiosis**: 1 diploid → 4 haploid; **meiosis I separates homologs (reductional), crossing over in prophase I; meiosis II separates sister chromatids**. Nondisjunction → trisomy (e.g., Down; risk ↑ with maternal age, mostly meiosis I).

5. Apoptosis

Apoptosis	Necrosis
ATP-dependent, programmed; single cells	Passive; groups of cells
Cell shrinks; pyknosis → karyorrhexis → karyolysis; eosinophilic cytoplasm; apoptotic bodies	Cell swells, membrane ruptures, contents leak
NO inflammation (membrane intact; PS "eat-me" → phagocytosis)	Inflammation
DNA laddering (internucleosomal cleavage, ~180 bp)	Random DNA degradation (smear)

- **Physiologic roles**: embryogenesis (interdigital webs, lumen formation), hormone withdrawal (endometrium, menstrual shedding), **thymic negative selection**, immune cell deletion, neutrophil turnover, cytotoxic T-cell killing, keeping tissue size constant.
- **Intrinsic (mitochondrial) pathway**: DNA damage, growth-factor withdrawal, stress → p53 → **pro-apoptotic Bax/Bak** form pores (BH3-only proteins Bim, Bad, PUMA, NOXA activate) → **cytochrome c release** → binds Apaf-1 → apoptosome → **caspase 9 (initiator)** → caspase 3/6/7 (executioners). **Bcl-2 and Bcl-xL are anti-apoptotic** (keep the outer membrane closed).
- **Extrinsic (death receptor) pathway**: **Fas ligand–Fas (CD95)** or TNF–TNFR1 → FADD adaptor → **caspase 8 (initiator)** → executioner caspases. Cytotoxic T cells also use **perforin + granzyme B** (activates caspases directly).
- IAPs/**survivin** inhibit caspases (overexpressed in tumors). Released Smac/DIABLO blocks IAPs.
- Defective Fas/FasL → **autoimmune lymphoproliferative syndrome** (failure to delete autoreactive lymphocytes).
- **BCL-2 overexpression: follicular lymphoma, t(14;18)** (BCL-2 moved next to Ig heavy-chain promoter) → cells don't die. CLL also overexpresses BCL-2 (venetoclax).
- **Detection**: **annexin V (binds externalized PS — early)**, TUNEL (DNA fragments), DNA laddering on gel, caspase assays, cytochrome c release.
- Too little apoptosis → cancer, autoimmunity. Too much → neurodegeneration (Alzheimer, Parkinson, ALS), ischemic injury, AIDS (CD4 loss).

6. Rapid-fire NBME pearls

- **I-cell disease = no M6P tag (GlcNAc-phosphotransferase) → enzymes in plasma.**
- **Zellweger (no peroxisomes), X-ALD (VLCFA, adrenal + demyelination), Refsum (phytanic acid).**
- **Ragged-red fibers + maternal inheritance + lactic acidosis = mitochondrial disease.**
- **Vincristine/vinblastine block assembly; paclitaxel blocks disassembly; colchicine for gout.**
- **Kartagener: dynein arm defect, situs inversus, infertility, bronchiectasis.**
- **Cytokeratin carcinoma, vimentin sarcoma, desmin muscle, GFAP glia.**
- **Intrinsic = cytochrome c / caspase 9; extrinsic = Fas / caspase 8; BCL-2 t(14;18) follicular lymphoma; annexin V detects PS.**
- **p53 → p21 → G1 arrest; Rb holds E2F; HPV E6 (p53), E7 (Rb).**

- COPII forward (ER → Golgi), COPI backward, clathrin to lysosomes/endocytosis.
- Cholesterol buffers fluidity; PS inner leaflet; cholera toxin binds GM1.